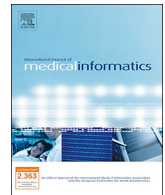




Contents lists available at ScienceDirect

International Journal of Medical Informatics

journal homepage: www.elsevier.com/locate/ijmedinf



Examining algorithmic biases in YouTube's recommendations of vaccine videos



Deena Abul-Fottouh^a, Melodie Yunju Song^{b,*}, Anatoliy Gruzd^c

^a Faculty of Information (iSchool), University of Toronto, 140 St. George Street, Toronto, Ontario, M5S 3G6, Canada

^b Social Media Lab, Ted Rogers School of Management, Ryerson University, 10 Dundas Street East #1002, Toronto, Ontario, M5B 2G9, Canada

^c Ted Rogers School of Information Technology Management, Ryerson University, 350 Victoria Street, Toronto, Ontario, M5B 2K3, Canada

ARTICLE INFO

Keywords:

Misinformation
Disinformation
Vaccination
YouTube
Network analysis
Social media
Recommender algorithm

ABSTRACT

Objective: This research examines how YouTube recommends vaccination-related videos.

Materials and methods: We used a social network analysis to evaluate how YouTube recommends vaccination related videos to its users.

Results: More pro-vaccine videos (64.75%) than anti-vaccine (19.98%) videos are on YouTube, with 15.27% of videos being neutral in sentiment. YouTube was more likely to recommend neutral and pro-vaccine videos than anti-vaccine videos. There is a homophily effect in which pro-vaccine videos were more likely to recommend other pro-vaccine videos than anti-vaccine ones, and vice versa.

Discussion: Compared to our prior study, the number of recommendations for pro-vaccine videos has significantly increased, suggesting that YouTube's demonization policy of harmful content and other changes to their recommender algorithm might have been effective in reducing the visibility of anti-vaccine videos. However, there are concerns that anti-vaccine videos are less likely to lead users to pro-vaccine videos due to the homophily effect observed in the recommendation network.

Conclusion: The study demonstrates the influence of YouTube's recommender systems on the types of vaccine information users discover on YouTube. We conclude with a general discussion of the importance of algorithmic transparency in how social media platforms like YouTube decide what content to feature and recommend to its users.

1. Introduction

YouTube is one of the most popular [1] but understudied social media platforms. It reaches 95% of the Internet population, with content available in 80 different languages [2]. Each month, over 2 billion unique users visit the site, and more than a billion hours of video are being watched each day [3]. YouTube is also one of the leading sources of health misinformation [4], and misinformation about vaccines in particular [5–7]. When searching for “vaccine safety” and “vaccines and children”, Basch and colleagues found that 65.5% of 87 videos in their sample were anti-vaccine [7]. Another study showed that 74% of vaccine videos on YouTube are anti-vaccine [8]. The above-mentioned small-scale studies (< 200 videos) highlight the growing prevalence of anti-vaccine videos on YouTube; however, more can be learned from examining a larger sample of YouTube videos. There is also a research gap when it comes to understanding the built-in biases of recommender algorithms that platforms like YouTube use to curate content in real-

time to millions of its users [9–11], and specifically their role in potentially spreading vaccine misinformation.

While the actual mechanism behind YouTube's recommender algorithm is unknown due to its proprietary nature; it is reported to rely on deep neural networks to learn from videos watched, click-through rates, average time spent on the video, and engagement (i.e., disliking or liking a video, commenting) to tailor video recommendations [12] with the goal of maximizing viewership, watch time, and user retention [13] in order to generate advertising revenue for YouTube. This suggests that the algorithm does not necessarily consider content accuracy in its recommendation decisions as long as the audience provides feedback that the content is welcomed. As a result, popular misinformation may be propagated because of existing algorithmic biases. This topic is especially important since previous research has shown that video recommendations are one of the main drivers of video views on YouTube [14,15]. From the perspective of interface affordance, there are 5 ways a user is recommended videos: recommendations

* Corresponding author.

E-mail addresses: deena.abulfottouh@utoronto.ca (D. Abul-Fottouh), yunju.song@ryerson.ca (M.Y. Song), gruzd@ryerson.ca (A. Gruzd).

<https://doi.org/10.1016/j.ijmedinf.2020.104175>

Received 8 August 2019; Received in revised form 29 April 2020; Accepted 9 May 2020

1386-5056/ © 2020 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

appearing after querying keywords in the search bar; suggested videos that are personalized based on user activities; a section showcasing new/unwatched videos of subscribed channels of the user; and a feed showing popular videos that are trending in a particular geographic region. In this work, we will only focus on recommended videos.

Our study is guided by two interrelated concepts: homophily and filter-bubble effects. Homophily effect refers to the notion that like-minded individuals (those who share the same beliefs, behaviours, and habits) tend to act in a similar way [16]. Following this principle, and since YouTube's recommender algorithm tends to recommend similar content to users with similar viewing habits, we expect that video recommendations would also be clustered around similar sentiments towards vaccination (pro- or anti-). If it happens, this may lead to the phenomenon called filter bubble [17], which occurs when a recommender system makes assumptions of user preferences based on prior collected information about that user, making it less likely that the user would be exposed to diverse perspectives. This is potentially dangerous because if users get "stuck" in a filter bubble of health misinformation, it may lead to the entrenchment of false beliefs shared by a group of homogenous individuals. Thus, we ask:

- 1 What is the prevalence of pro-, neutral- and anti-vaccine videos on YouTube?
- 2 In the network of vaccine-related video recommendations, does the sentiment of the video (anti-vaccine or pro-vaccine) influence the likelihood of a video being recommended? In other words, are anti-vaccine videos more likely to be recommended than pro-vaccine videos?
- 3 Do we observe a homophily effect among pro- and anti-vaccine videos? For example, do pro-vaccine videos tend to recommend more pro-vaccine videos and vice versa?

2. Material and methods

2.1. Data collection

Following our previous methodology [18], we used YouTube Data Tools [19] to retrieve YouTube videos related to vaccination and discover a network of recommended videos, in which videos are linked based on YouTube's recommender algorithm. Specifically, we used "immuni* OR vaccin* OR vaxx*" as a search query to retrieve the initial list of 250 videos with English titles (seed videos) ranked by YouTube as most relevant to the given search query. For each video on this list, YouTube Data Tools retrieved up to 50 video recommendations provided by YouTube API. In total, the resulting dataset contained 8425 videos and associated metadata, including comment count, view count, dislike count, like count, dislike/like ratio, published date, channel ID and channel title. The dataset was retrieved on April 9, 2019.

Since our research only focuses on YouTube videos related to vaccination, we excluded any videos which titles did not contain relevant vaccination-related keywords (immuni*, vaxx*, vaccin*). In addition, we excluded English titled videos with non-English content ("Immunisations vaccinations in Victoria Urdu"), and irrelevant videos on topics such as animal vaccination ("Administering Vaccines - Canine"), actuarial exam preparation videos ("Immunizing a Bond Portfolio"), or exam preparation videos ("FSc Biology Book1 CH 6 LEC 12: Immunization and Vaccination"). We also excluded videos that were not available at the time of human coding. The final dataset consists of 2122 videos.

2.2. Vaccine sentiment and video affiliation coding

One of the authors with expertise in public health watched all selected vaccine-related videos to determine their sentiment towards vaccination and categorize them into three types: pro-vaccine, neutral, and anti-vaccine. Pro-vaccine videos were in support of immunization,

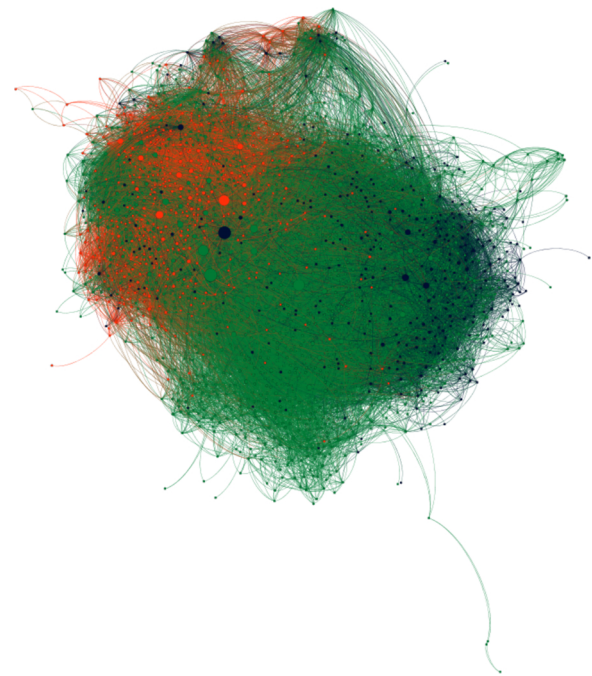


Fig. 1. Network of YouTube's vaccine-related videos. Nodes represent videos and connections represent YouTube's recommendation of videos. Pro-vaccine videos are shown in green, anti-vaccine videos in red, neutral videos in black. The colours show communities of pro-vaccine versus anti-vaccine videos. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

while anti-vaccine videos featured attitudes of refusal or rejection towards vaccines. Neutral sentiment videos were neither supportive of nor opposed to vaccines (e.g., news media presenting two sides of the "debate"). Initially, we recruited a second coder to validate the first round of coding. However, since the interrater agreement between the coders on the initial set of videos coded by both coders ($n = 1840$) was only considered 'moderate' (Cohen's kappa of 0.43) [20], we recruited two additional coders. A third and fourth coder reviewed diverging responses until a consensus on video sentiment was reached between at least two out of four coders for all 2122 vaccine-related videos in the sample. The resulting vaccination sentiment codes were used to answer RQ1.

In addition to the sentiment coding, we also determined whether a video was posted by a health-related organization (e.g., hospital, public health organization, public health unit), and whether a video was posted by or featured individuals who are health professionals (e.g., clinician, physician, nurse).

Fig. 1 shows the resulting network of 2122 recommended videos. In this network, each node represents a single video with vaccination content, and the connections represent YouTube's recommendation of videos. Pro-vaccine videos are shown in green, anti-vaccine videos in red, neutral videos in black.

2.3. Social network analysis

To analyse the retrieved network of 2122 recommended videos, we turned to social network analysis and, specifically, a Latent Order Logistic Model (LOLOG) [21]. LOLOG models belong to exponential family probability models that offer a fully general framework capable of describing any probability distribution over network configurations. We chose the LOLOG model as it is a robust network analysis methodology. First, this type of probability model is designed to account for processes behind network growth over time, such as those commonly observed in online networks (ibid). Second, LOLOG models are similar

to the popular Exponential-Family Random Graph Models (ERGM) [23], which rely on Monte Carlo methods and variational inference to describe probability distributions in large networks. However, the advantage LOLOG models have over ERGMs is that they can model scale-free property of YouTube networks, while avoiding problems of model degeneracy. Third, LOLOG models are easily interpreted in the same way logistic regression models are. For these reasons, we use LOLOG models in this research to predict the probability of tie formation among vaccine-related YouTube videos.

Using LOLOG, we tested whether videos tend to recommend other videos or be recommended based on their sentiment towards vaccination (RQ2), and if videos with the same sentiment tend to recommend each other (RQ3). The dependent variable for both RQ2 and RQ3 is the probability of tie formation among videos in the recommendation network. To answer RQ2, we modelled the network's overall propensity to form ties by adding the sentiment of the video towards vaccination as the main independent variable in the model. This is a categorical variable with anti-vaccine videos as the baseline. To answer RQ3, the main independent variable was whether videos connected by a tie share the same sentiment. We also examined whether considerable reciprocity among videos existed. Reciprocity indicates whether, if a video recommends another video, the latter tends to recommend it back. Additionally, we included the within-group edges among videos produced by health-related organizations or healthcare professionals as control variables to the model.

Due to the nature of online networks that usually follow power-law distributions and its tendency to have skewed in-degree (the number of incoming connections) and out-degree distributions (the number of outgoing connections), we controlled for in-degree values between 0–4, out-degree values between 0–4, and out-degrees of two stars and three stars in order to model the skewness of the in-degree and out-degree distributions in the observed network. All attributes related to video viewership such as the number of dislikes, likes, comment count, or view counts did not have any significant effect on video recommendations; thus, they were not included in the final model.

3. Results

3.1. Pro-vaccine videos are more prevalent than anti-vaccine videos

To answer RQ1, our search returned nearly three times more pro-vaccine videos than anti-vaccine. Among the 2122 vaccine-related videos, 1374 videos were pro-vaccine (64.75%), 424 videos were anti-vaccine (19.98%), and 324 videos were neutral (15.27%). 1568 (73.89%) out of the 2122 videos were neither posted by nor featured healthcare professionals, suggesting that the vaccine related content on YouTube is mostly dominated by non-healthcare professionals. It is interesting to note that our previous study conducted using the same query terms and the same data collection tool in 2016 yielded almost an opposite result showing the prevalence of anti-vaccine over pro-vaccine videos then [18].

3.2. Effect of video sentiment on video recommendations

Table 1 summarizes the results of the LOLOG model, including: a vector of graph statistics $g(y)$ for the network, the set of coefficients θ for the statistics, the corresponding standard error and p-value. The probability of observing a recommendation tie between two videos (the dependent variable in the model) is calculated based on the set of graph statistics $g(y)$. Since the LOLOG model is an exponential family model, the probability of tie formation is formulated as a logistic regression, using the change in the graph statistics as predictors [21]. The coefficients θ are thus interpreted in the same way as in a logistic regression.

To answer RQ2, we included the video sentiment variable as independent variables (rows 3–6 in Table 1). Video sentiment is a categorical variable where anti-vaccine videos is the baseline category. We

distinguish between videos having the propensity to receive (incoming ties) or send recommendations (outgoing ties). To answer RQ3, we included the “Sentiment (Match)” variable as an independent variable (row 7 in Table 1). This is a binary categorical variable with the baseline category being two videos not sharing the same sentiment.

We also included various control variables: whether there are reciprocal recommendations among videos posted by health-related organizations (row 8) or healthcare professionals (row 9), or whether videos recommend each other in general (Reciprocity in row 10). Finally, to account for the possibility that videos might be recommended because of their popularity, we controlled for degree centrality (the number of incoming and outgoing ties, rows 11–20) and node centralization (rows 21–22).

Results show that pro-vaccine videos were more likely to be recommended by YouTube than anti-vaccine videos, since the coefficient for this variable is statistically significant (see row 4 in Table 1, $\theta = 0.44$, $p < 0.001$). Likewise, pro-vaccine videos were more likely to generate more recommendations than anti-vaccine videos (see row 6 in Table 1, $\theta = 0.17$, $p < 0.001$).

3.3. Homophily effect of videos by vaccine sentiment

In response to RQ3, the LOLOG model suggests that there is a homophily effect based on video sentiment. Pro-vaccine videos are more likely to be recommended by other pro-vaccine videos and the same applies for anti-vaccine videos ($p < .001$). The coefficient θ for this model parameter (Sentiment Match) is 0.90 suggesting that the odds of forming a tie between videos with the same sentiment is $\exp(0.90) = 2.5$ times higher than forming a tie between videos of different sentiments (see row 7 in Table 1). Furthermore, the observed homophily effect is further reinforced by the reciprocal nature of recommendations, meaning that if YouTube recommends video B based on video A, it is more likely than by chance alone that video B will also lead to video A (see row 10 in Table 1).

To validate the resulting model, we ran 1000 simulations to compare various statistics between the simulated networks and the observed network [24]. Fig. 2 shows the goodness of fit for the degree distributions (in-degree and out-degree) and edge-wise shared partner (ESP) distribution of simulations of statistics from the fitted model compared to the observed network. These parameters were used to see how well the model fits patterns that are not explicitly represented by the terms in the model. The red line shows the observed network and the black lines are the simulated ones. As the black and red lines overlap and generally follow each other's pattern, the model shows a good fit with regards to these metrics.

4. Discussion

We sought to understand how vaccine-related videos are recommended on YouTube. In contrast to our previous study where we found that 65% of vaccine-related videos sampled in 2016 were anti-vaccine [18]; in the current study, 64.75% of vaccine-related videos were pro-vaccine.

Second, there is a trend that pro-vaccine videos are more likely to be recommended in comparison to anti-vaccine videos ($p < 0.001$). Both of these findings may be a result of recent changes to YouTube's recommender system in 2019 to demonetize anti-vaccine videos and make pro-vaccine content from authoritative contributors easier to locate on YouTube [25,26].

Third, the observed homophily based on video sentiment suggests that there is a filter-bubble effect among vaccine-related videos. While this finding may be an indication that viewers of pro-vaccine videos are not exposed to vaccine misinformation, there is the downside that viewers of anti-vaccine videos are also less likely to be exposed to credible health information about vaccination. Viewers who are “stuck” in an anti-vaccine filter bubble may not realize that videos

Table 1
LOLOG model of YouTube vaccine-related video recommendation network.

		Observed Graph Statistics g(y)	Estimated Model Coefficients θ	Std. Error	p-value
1	Edges	36761.00	3.35	0.14	0.000
2	Preferential Attachment	NA*	1.22	0.01	0.000
3	Incoming Ties-Neutral	8738	0.38	0.07	0.000
4	Incoming Ties-Pro-Vaccine	5726	0.44	0.08	0.000
5	Outgoing Ties-Neutral	8378	0.27	0.03	0.000
6	Outgoing Ties-Pro-Vaccine	5072	0.17	0.04	0.000
7	Sentiment (Match)	22,763	0.90	0.02	0.000
8	Health-related Organization (Match)	27,409	0.25	0.05	0.000
9	Healthcare Professional (Match)	23,465	0.07	0.04	0.112
10	Reciprocity	4183	3.64	0.05	0.000
11	In-degree0	53	1.69	0.33	0.000
12	In-degree1	268	1.15	0.26	0.000
13	In-degree2	217	0.29	0.22	0.191
14	In-degree3	167	-0.08	0.16	0.608
15	In-degree4	142	-0.12	0.11	0.276
16	Out-degree0	149	0.98	0.52	0.058
17	Out-degree1	56	-0.26	0.42	0.527
18	Out-degree2	55	-0.35	0.33	0.299
19	Out-degree3	47	-0.33	0.24	0.180
20	Out-degree4	54	-0.11	0.16	0.488
21	Out-star2	474,018	0.16	0.01	0.000
22	Out-star3	4,908,298	-0.01	0.00	0.000

* Preferential Attachment accounts for the unobserved order in which dyads were added to the network, hence the presence of NA (not available) under observed statistics [22].

recommended to them that are likely to increase viewership retention and video shareability [9,27,28], do not necessarily contain credible health information.

Related to our finding of the filter-bubble effect, we also found that videos by healthcare organizations are significantly more likely to be linked to each other than to videos posted by non-healthcare entities (see row 7 in Table 1). This supports that creating pro-vaccine videos alone might not be an effective method to disseminate credible information about vaccination as it is less likely to be accessed by anti-vaccine leaning viewers. Thus, policies and funding to support pro-vaccination messages such as the production of documentaries, news-casts, and livestreams by reputable institutions are encouraged to integrate public health messages on YouTube; however, health communication on social media should be more than just providing credible information. It is also about understanding what types of videos will be watched and recommended by platforms like YouTube. To engage meaningfully on social media is to create content that is captivating and relatable, but also to go where their intended audience is, which in this case may include commenting sections of anti-vaccine videos. Continuous engagement on platforms like YouTube is particularly important because health professionals are often viewed as mediators who increase public's trust and satisfaction in health-seeking on social media [29].

Finally, while the current case study focused on examining vaccine-related videos and recommendations on YouTube, the findings also highlight the need for algorithmic transparency on social media more broadly, especially when it comes to recommending health-related information. Public health institutions and professionals ought to work with developers of social media platforms to be able to audit the nature of artificial intelligence-driven recommendations. Such collaboration will help to better understand and address any inherent biases built into either recommendation models or interface affordances, with the overall goal to minimize the chance that some users would be "stuck" in a filter bubble of misinformation.

5. Conclusion

This work sheds light on an important topic of how YouTube recommends vaccine-related videos and the role of inherent algorithmic biases in this process. The importance of this topic also comes from the

trend that the platform is increasingly being used in low- and middle-income countries [30]; as a result, the presence of health misinformation on platforms like YouTube could potentially lead to negative health behaviours, possibly resulting in sustained or periodic outbreaks in countries with low immunization rates and low access to vaccines.

While the study found a higher prevalence of pro-vaccine videos on YouTube, the presence of a filter bubble among anti-vaccine videos calls for technology companies to design recommender systems that could account for the quality and credibility of video content, especially when it comes to health information. At the same time, technology companies should be more transparent on how recommender systems operate. Monitoring and safeguarding the quality of health information online has been inhibited by the proprietary nature of the recommender systems on social media platforms, as their incremental modifications are often unknown to its users.

We conclude this paper with a few recommendations for future work. First, as described in the methods section, to ensure the inclusion of relevant videos about vaccination, the study excluded non-English videos and those which titles did not contain one of the selection keywords (immuni*, vaxx* or vaccin*). Future work should validate the results by including the analysis of videos in other languages and videos which may not necessarily use vaccination-related keywords. Second, besides vaccine sentiment of a video, identity of the video uploader and their affiliations, other properties may also influence how YouTube's recommender algorithm operates and should be examined, including the number of channel subscribers, video duration, and when it was uploaded. Developing experimental designs to understand how online information influence vaccine strategies [31], and employing methods leveraging intrinsic *user-item-query* connections such as the "ternary interactive item recommendation (TIIREC)" may assist in better video recommendations [32]. Third, while we relied on a manual content analysis of videos in this study, considering improvements in accuracy of video analysis techniques [33], they can be applied to scale up this work in the future. Finally, our analysis of YouTube's recommender algorithm is limited since we collected related videos by using YouTube's API. In practice, users receive recommendations based on a number of factors unique to them, including their past viewing history and location. Future work is needed to validate our findings by sampling video recommendations based on the actual accounts on YouTube.

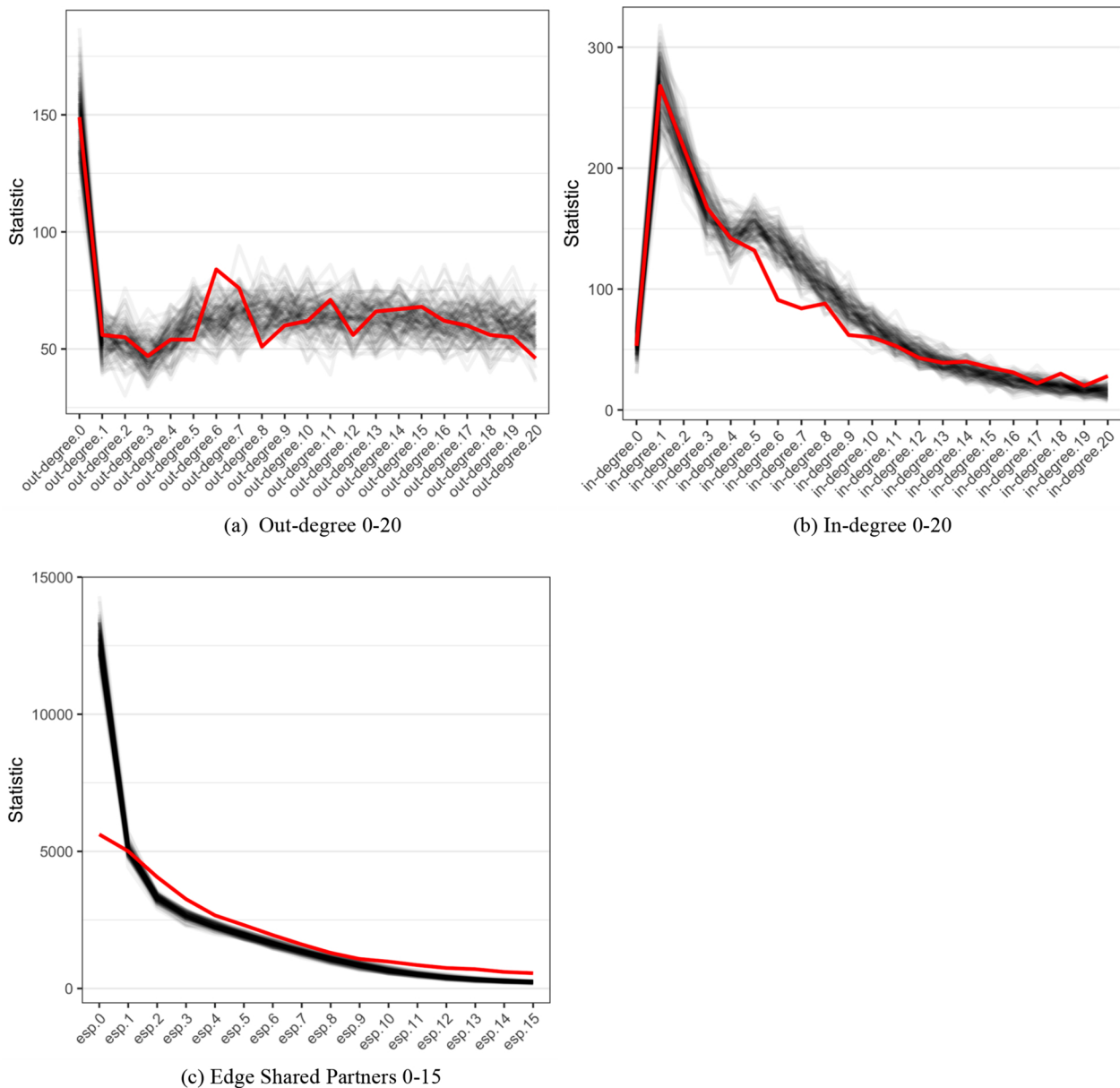


Fig. 2. The Out-Degree (a), In-Degree (b), and Edge Shared Partners (c) distributions from 1000 simulated networks (indicated by the black line) from the fitted LOLOG are compared to the observed video recommendations network (indicated by the red lines). The black and red lines follow each other indicating a good fit between the simulated model and the observed network. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

Funding sources

This research is supported in part through a five-year initiative on “Social Media Data Stewardship” funded by the Canada Research Chairs program (2015–2020; Principal Investigator: Gruz, A.).

Declaration of competing interest

None.

Summary table

What was already known on the topic:
• YouTube is one of the most popular but understudied social media platforms. • YouTube is also one of the leading sources of health

misinformation, and misinformation about vaccines in particular.
• YouTube’s recommender algorithm is unknown due to its proprietary nature, but is a major mechanism for users to discover new content on the platform.
What this study added to our knowledge:
• The number of recommendations for pro-vaccine videos has increased dramatically since a previous study in 2016, suggesting that YouTube’s demonization policy of harmful content and other changes to their recommender algorithm may have lowered the visibility of anti-vaccine videos. • Anti-vaccine videos are less likely to lead users to pro-vaccine videos due to the homophily effect observed in the recommendation network.

CRediT authorship contribution statement

Deena Abul-Fottouh: Formal analysis, Validation, Investigation, Visualisation, Methodology, Writing - original draft, Writing - review & editing. **Melodie Yunju Song:** Data curation, Project administration, Methodology, Writing - original draft, Writing - review & editing. **Anatoliy Gruz:** Conceptualization, Funding acquisition, Supervision, Validation, Methodology, Writing - original draft, Writing - review & editing.

Acknowledgements

The authors would like to thank Ian Fellows for his help with the LOLOG model fitting procedure and members of the Social Media Lab at Ryerson University, and specifically: Philip Mai and Lilach Dahoah Halevi for their invaluable feedback on this project, and Rebeca Chacon Cabrera and Donald Patterson for their help with the manual content analysis of videos.

Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.ijmedinf.2020.104175>.

References

- [1] J. Burgess, J. Green, YouTube: Online Video and Participatory Culture, John Wiley & Sons, 2018.
- [2] YouTube Policies, YouTube, 2019 (accessed January 28, 2019), <https://www.youtube.com/yt/about/policies/#community-guidelines>.
- [3] T. Spangler, YouTube Now Has 2 Billion Monthly Users, Who Watch 250 Million Hours on TV Screens Daily, Variety, (2019) (accessed August 6, 2019), <https://variety.com/2019/digital/news/youtube-2-billion-users-tv-screen-watch-time-hours-1203204267/>.
- [4] K.C. Madathil, A.J. Rivera-Rodriguez, J.S. Greenstein, A.K. Gramopadhye, Healthcare information on YouTube: a systematic review, Health Inform. J. 21 (2015) 173–194, <https://doi.org/10.1177/1460458213512220>.
- [5] J. Keelan, V. Pavri-Garcia, G. Tomlinson, K. Wilson, YouTube as source of information on immunization: a content analysis, JAMA 298 (2007) 2482, <https://doi.org/10.1001/jama.298.21.2482>.
- [6] R. Briones, X. Nan, K. Madden, L. Waks, When vaccines go viral: an analysis of HPV vaccine coverage on YouTube, Health Commun. 27 (2012) 478–485, <https://doi.org/10.1080/10410236.2011.610258>.
- [7] C.H. Basch, P. Zyburt, R. Reeves, C.E. Basch, What do popular YouTube videos say about vaccines? Child Care Hlth. Dev. 43 (2017) 499–503, <https://doi.org/10.1111/cch.12442>.
- [8] A. Venkatraman, N. Garg, N. Kumar, Greater freedom of speech on Web 2.0 correlates with dominance of views linking vaccines to autism, Vaccine. 33 (2015) 1422–1425, <https://doi.org/10.1016/j.vaccine.2015.01.078>.
- [9] Z. Tufekci, Algorithmic harms beyond Facebook and Google: emergent challenges of computational agency, Colorado Technol. Law J. 13 (2015) 203.
- [10] A. Caliskan, J.J. Bryson, A. Narayanan, Semantics derived automatically from language corpora contain human-like biases, Science 356 (2017) 183–186, <https://doi.org/10.1126/science.aal4230>.
- [11] H. Cramer, J. Garcia-Gathright, A. Springer, S. Reddy, Assessing and addressing algorithmic bias in practice, Interactions 25 (2018) 58–63, <https://doi.org/10.1145/3278156>.
- [12] P. Covington, J. Adams, E. Sargin, Deep neural networks for youtube recommendations, Proceedings of the 10th ACM Conference on Recommender Systems, ACM (2016) 191–198.
- [13] A. Beutel, P. Covington, S. Jain, C. Xu, J. Li, V. Gatto, E.H. Chi, Latent cross: making use of context in recurrent recommender systems, Proceedings of the Eleventh ACM International Conference on Web Search and Data Mining, ACM (2018) 46–54, <https://doi.org/10.1145/3159652.3159727>.
- [14] R. Zhou, S. Khemmarat, L. Gao, The impact of YouTube recommendation system on video views, Proceedings of the 10th ACM SIGCOMM Conference on Internet Measurement, ACM (2010) 404–410, <https://doi.org/10.1145/1879141.1879193>.
- [15] R. Zhou, S. Khemmarat, L. Gao, J. Wan, J. Zhang, How YouTube videos are discovered and its impact on video views, Multimed. Tools Appl. 75 (2016) 6035–6058 doi: 10/1007/s11042-015-3206-0.
- [16] M. McPherson, L. Smith-Lovin, J.M. Cook, Birds of a feather: homophily, Annu. Rev. Sociol. 27 (2001) 415–444, <https://doi.org/10.1146/annurev.soc.27.1.415>.
- [17] E. Pariser, The Filter Bubble: How the New Personalized Web Is Changing What We Read and How We Think, Penguin, 2011.
- [18] M. Song, A. Gruz, Examining sentiments and popularity of pro- and anti-vaccination videos on YouTube, Proceedings of the 8th International Conference on Social Media & Society, ACM, New York, NY, USA, 2017, pp. 17:1–17:8, <https://doi.org/10.1145/3097286.3097303>.
- [19] B. Rieder, YouTube Data Tools. Computer software. Vers. 1.0. (2015) <https://tools.digitalmethods.net/netvizz/youtube>.
- [20] M.L. McHugh, Interrater Reliability: the Kappa Statistic, Biochemia Medica 22 (2012) 276–282.
- [21] I.E. Fellows, A New Generative Statistical Model for Graphs: The Latent Order Logistic (LOLOG) Model, ArXiv:1804.04583 [Cs, Stat], (2018) (accessed July 18, 2019), <http://arxiv.org/abs/1804.04583>.
- [22] The Statnet Development Team, An Introduction to Latent Order Logistic (LOLOG) Network Models, R Foundation for Statistical Computing, 2019, <https://cran.r-project.org/web/packages/lolog/vignettes/lolog-introduction.pdf>.
- [23] D.R. Hunter, M.S. Handcock, Inference in curved exponential family models for networks, J. Comput. Graph. Stat. 15 (2006) 565–583, <https://doi.org/10.1198/106186006X133069>.
- [24] D.R. Hunter, M.S. Handcock, C.T. Butts, S.M. Goodreau, M. Morris, Ergm: a package to fit, simulate and diagnose exponential-family models for networks, J. Stat. Softw. 24 (2008) nihpa54860.
- [25] R. Becker, New Research Targets Anti-vaccination Comments on Facebook, The Verge, 2019 (accessed March 29, 2019), <https://www.theverge.com/2019/3/21/18275371/facebook-anti-vaccination-vaxx-trust-media-measles-mmr>.
- [26] BBC, YouTube Takes Ads Off “anti-vax” Videos, BBC News, 2019 (accessed October 9, 2019), <https://www.bbc.com/news/technology-47357252>.
- [27] S. Syed-Abdul, L. Fernandez-Luque, W.-S. Jian, Y.-C. Li, S. Crain, M.-H. Hsu, Y.-C. Wang, D. Khandregzen, E. Chuluunbaatar, P.A. Nguyen, Misleading health-related information promoted through video-based social media: anorexia on YouTube, J. Med. Internet Res. 15 (2013), <https://doi.org/10.2196/jmir.2237>.
- [28] S. Bishop, Anxiety, panic and self-optimization: inequalities and the YouTube algorithm, Convergence 24 (2018) 69–84, <https://doi.org/10.1177/1354856517736978>.
- [29] T. Wu, Z. Deng, Z. Feng, D.J. Gaskin, D. Zhang, R. Wang, The effect of doctor-consumer interaction on social media on consumers' health behaviors: cross-sectional study, J. Med. Internet Res. 20 (2018) e73, <https://doi.org/10.2196/jmir.9003>.
- [30] E. Hagg, V.S. Dahinten, L.M. Currie, The emerging use of social media for health-related purposes in low and middle-income countries: a scoping review, Int. J. Med. Inform. 115 (2018) 92–105, <https://doi.org/10.1016/j.ijmedinf.2018.04.010>.
- [31] X.X. Zhan, C. Liu, G. Zhou, Z.K. Zhang, G.Q. Sun, J.J. Zhu, Z. Jin, Coupling dynamics of epidemic spreading and information diffusion on complex networks, Appl. Math. Comput. 332 (2018) 437–448.
- [32] L. Yu, J. Huang, G. Zhou, L. Chuang, Z.-K. Zhang, TIIRC: a Tensor approach for tag-driven item recommendation with sparse user generated content, Inf. Sci. 411 (2017) 122–135, <https://doi.org/10.1016/j.ins.2017.05.025>.
- [33] S. Poria, N. Majumder, R. Mihalcea, E. Hovy, Emotion Recognition in Conversation: Research Challenges, Datasets, and Recent Advances, ArXiv, 2019 (accessed July 23, 2019), 1905.02947 [Cs], <http://arxiv.org/abs/1905.02947>.

Update

International Journal of Medical Informatics

Volume 148, Issue , April 2021, Page

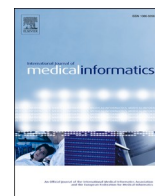
DOI: <https://doi.org/10.1016/j.ijmedinf.2021.104385>



Contents lists available at ScienceDirect

International Journal of Medical Informatics

journal homepage: www.elsevier.com/locate/ijmedinf



Corrigendum

Corrigendum to “Examining algorithmic biases in YouTube’s recommendations of vaccine videos” [Int. J. Med. Inf. 140 (2020) 104175]

Deena Abul-Fottouh^a, Melodie Yunju Song^b, Anatoliy Gruzd^{c,*}

^a Faculty of Information (iSchool), University of Toronto, 140 St. George Street, Toronto, Ontario, M5S 3G6, Canada

^b Social Media Lab, Ted Rogers School of Management, Ryerson University, 10 Dundas Street East #1002, Toronto, Ontario, M5B 2G9, Canada

^c Ted Rogers School of Information Technology Management, Ryerson University, 350 Victoria Street, Toronto, Ontario, M5B 2K3, Canada

The authors regret to notify about an unfortunate typo in the word “demonetization” in two places: in the abstract and in the summary table at the end of the published manuscript. In particular, the word “demonization” in “YouTube’s demonization policy” should be replaced with “YouTube’s demonetization policy”.

In addition, there is a further typo in the Introduction that currently states “From the perspective of interface affordance, there are 5 ways...” and should read “From the perspective of interface affordance, there are 4 ways...”.

The authors would like to apologise for any inconvenience caused.

DOI of original article: <https://doi.org/10.1016/j.ijmedinf.2020.104175>.

* Corresponding author.

E-mail addresses: deena.abulfottouh@utoronto.ca (D. Abul-Fottouh), yunju.song@ryerson.ca (M.Y. Song), gruzd@ryerson.ca (A. Gruzd).

<https://doi.org/10.1016/j.ijmedinf.2021.104385>

Available online 27 January 2021

1386-5056/© 2021 The Author(s).

Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license

(<http://creativecommons.org/licenses/by-nc-nd/4.0/>).